

CHESTER CATAWBA, SC, USA

WMO: 720599

Lat: 34.783N

Lon: 81.200W

Elev: 657

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00192

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.3	27.0	2.6	6.4	29.2	9.2	9.0	33.3	17.0	52.4	15.5	46.8	3.2	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	94.9	74.6	91.3	73.7	90.3	73.5	80.9	86.7	78.3	84.6	76.8	84.4	4.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	155.2	84.6	77.0	144.1	80.9	74.6	132.7	78.3	45.1	86.0	42.2	85.0	40.6	86.3	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.8	11.1	16.9	97.7	4.4	3.8	13.8	100.5	11.2	102.7	8.7	104.9	5.5	107.7
			15.5	79.9	4.1	2.7	12.6	81.9	10.2	83.5	7.9	85.0	4.9	87.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	42.4	47.0	52.2	62.3	70.6	77.7	80.0	78.4	74.2	62.4	51.4	46.5
	DBStd	15.08	9.65	9.17	9.05	7.30	6.17	4.15	3.42	3.34	5.24	7.61	7.15	8.83
	HDD50	764	274	154	84	5	0	0	0	0	0	5	66	176
	HDD65	2909	702	508	408	134	28	0	0	0	6	138	410	575
	CDD50	5205	37	70	151	376	638	831	931	882	725	388	108	68
	CDD65	1875	1	3	10	55	200	382	466	417	281	56	2	2
	CDH74	16374	1	15	102	526	1602	3504	4430	3469	2272	444	8	1
	CDH80	6662	0	0	9	102	537	1529	2048	1447	886	104	0	0
(14) Wind	WSAvg	4.2	4.4	4.8	5.3	5.2	4.2	3.5	3.5	3.1	4.1	4.3	4.4	4.0
(15) Precipitation	PrecAvg	43.9	3.9	3.5	4.4	3.4	2.9	4.2	3.6	4.3	3.6	3.5	3.2	3.4
	PrecMax	55.0	7.0	6.0	8.3	7.9	7.6	9.1	7.7	9.7	9.4	12.6	6.9	7.2
	PrecMin	28.6	0.9	1.4	1.2	1.3	1.0	0.7	1.4	0.5	0.0	0.0	0.4	1.3
	PrecStd	7.0	1.7	1.3	2.0	1.9	1.5	2.2	1.4	2.4	2.3	2.7	1.8	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	76.9	81.3	86.3	91.2	97.4	98.7	95.1	93.1	88.4	74.5	72.2
		MCWB	61.3	62.8	61.2	66.2	69.6	74.0	75.1	74.8	71.7	70.8	62.0	65.0
	2%	DB	65.6	71.9	77.2	82.2	89.6	93.4	95.3	91.3	90.6	81.8	71.7	68.3
		MCWB	58.0	59.3	60.0	65.0	69.1	74.5	75.4	74.6	71.8	68.4	60.1	63.6
	5%	DB	62.7	67.7	72.3	80.8	86.1	90.6	91.5	90.2	88.4	79.1	68.0	64.0
		MCWB	56.3	56.4	58.0	64.1	69.2	73.7	74.2	74.6	71.5	65.6	60.3	60.2
	10%	DB	58.6	63.2	68.0	77.0	82.1	88.3	90.3	88.1	86.0	74.9	63.9	61.2
		MCWB	52.6	55.1	56.7	62.0	68.4	73.0	74.3	74.2	70.8	65.0	57.5	57.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	64.4	65.3	71.0	76.5	82.6	83.3	83.4	77.5	74.8	67.9	69.0
		MCDB	68.0	70.0	73.4	79.5	83.2	91.1	92.6	88.7	87.9	77.8	68.6	69.8
	2%	WB	61.7	62.7	63.2	69.1	73.7	79.9	80.3	80.8	75.8	72.7	64.4	65.7
		MCDB	63.5	67.6	69.8	75.7	81.0	86.2	86.5	84.2	85.0	76.5	67.2	68.0
	5%	WB	57.4	59.9	61.3	67.4	72.1	77.2	77.8	77.7	74.0	70.6	63.0	62.5
		MCDB	61.6	65.6	67.5	75.1	79.1	83.5	86.1	83.3	81.0	75.4	65.3	63.5
	10%	WB	53.3	56.1	59.2	65.7	70.7	74.8	76.4	76.3	72.7	68.2	59.8	57.8
		MCDB	55.6	61.3	65.9	74.1	77.4	83.6	85.2	83.0	79.2	73.7	63.1	61.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	19.8	21.2	22.8	20.4	19.9	19.4	18.4	20.0	21.1	20.7	17.3
		MCDBR	23.8	24.1	26.6	25.7	24.5	22.8	22.2	21.1	23.7	23.5	24.1	19.0
		MCWBR	16.0	13.6	13.1	10.7	8.5	7.4	6.5	6.7	7.6	10.4	15.0	12.8
		MCDBR	19.4	20.0	21.3	20.8	18.9	20.1	19.3	17.2	19.8	18.1	17.6	14.2
	5% WB	MCWBR	16.1	13.5	13.6	10.5	8.9	9.5	7.9	8.0	8.1	10.5	14.3	12.3
(40) Clear-Sky Solar Irradiance	taub		0.316	0.325	0.363	0.406	0.462	0.488	0.539	0.518	0.435	0.389	0.338	0.317
	taud		2.498	2.470	2.393	2.264	2.149	2.120	1.999	2.044	2.260	2.364	2.468	2.527
	Ebn at Noon		283	292	288	279	264	256	242	245	261	265	272	276
	Edh at Noon		27	30	36	43	49	50	56	53	40	34	27	24
(44) All-Sky Solar Radiation	RadAvg		851	1053	1394	1744	1928	2038	1909	1748	1495	1236	939	734
	RadStd		46	118	102	136	157	146	124	83	131	150	77	69

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+315	+214

Nomenclature: See separate page